

DHRUV BHAGAT

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SUMMARY

Computer Science undergraduate with hands-on experience in Python, backend development, REST APIs, databases, and cloud-based applications. Built scalable software systems using Python, FastAPI, Node.js, Express.js, MongoDB, PostgreSQL, Docker, and AWS. Strong foundation in OOP, DSA, DBMS, Operating Systems, and Computer Networks, with practical experience integrating AI capabilities into production-oriented applications.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Core CS: OOP, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks

Backend: FastAPI, Node.js, Express.js, REST APIs, Async Programming

Frontend: React.js, Tailwind CSS

Databases: MongoDB, PostgreSQL, Redis, ChromaDB

Cloud & DevOps: AWS EC2, Docker, Docker Compose, Git, GitHub

AI/ML: Machine Learning, PyTorch, Computer Vision, LLMs, Generative AI, RAG, Prompt Engineering

AI Tools: LangChain, Hugging Face Embeddings, n8n, NVIDIA NIM, Groq API

EXPERIENCE

AI for Sustainability Intern | 1M1B, AICTE & IBM SkillsBuild

May 2026 – June 2026

- Contributed to the development of an AI-powered waste classification application as part of a cross-functional development team.
- Developed and integrated 11 REST APIs for AI inference, analytics, data processing, and application workflows.
- Managed assigned development tasks and timelines while coordinating with AI, backend, and frontend team members to meet project milestones.
- Contributed to testing, integration, and project delivery while improving task prioritization, team coordination, and adaptability to new technologies.

PROJECTS

Multi-Source Enterprise RAG System

[GitHub](#)

- Built a multi-tenant document intelligence platform using Python, FastAPI, PostgreSQL, and ChromaDB, supporting document retrieval and citation-backed responses.
- Developed provider-agnostic AI interfaces using OOP and abstract base classes, integrating multiple LLM and embedding providers with semantic retrieval.
- Implemented asynchronous document processing and streaming responses using async/await, generators, Redis workers, and Server-Sent Events (SSE).

EcoSort AI

[GitHub](#)

- Developed a waste classification system using PyTorch and EfficientNet-B0 to classify waste across 10 categories.
- Built FastAPI inference services with image uploads, JWT authentication, and REST APIs for real-time AI predictions.
- Containerized frontend, backend, ML inference, and database services using Docker Compose.

End to End E-Commerce Churn Intelligence System

[GitHub](#)

- Built a customer churn prediction platform using Python, FastAPI, MongoDB, and Random Forest (Scikit-learn) supporting customer risk analysis and prediction workflows.
- Developed ETL and data ingestion pipelines with data preprocessing, model inference, state management, and database operations.
- Implemented backend workflows using REST APIs, structured logging, exception handling, type hints, and JSON-based data processing for reliable application execution.

EDUCATION

Chandubhai S. Patel Institute of Technology (CSPIT), Charusat University

8.71/10

Bachelor of Technology (B.Tech) – Computer Science and Engineering

CERTIFICATIONS

- AWS Academy Graduate – Cloud Developing (40 hours)
- NPTEL – Design and Analysis of Algorithms
- Machine Learning with Python – Coursera

